



FINAL EXAM BRIDGING SEMESTER II, SESSION 2020/2021

COURSE CODE	:	IFC 1024
COURSE NAME	:	CHEMISTRY
PROGRAMME	:	BRIDGING SPACE
DURATION	:	3 HOURS
DATE	:	06 JULY 2021

INSTRUCTION TO CANDIDATES:

1. DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.
2. ANSWER ALL QUESTIONS.

WARNING!

Students caught copying/cheating during the examination will be liable for disciplinary actions and UTMSPACE may recommend the student to be expelled from the study.

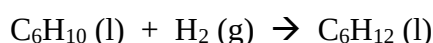
This booklet consists of (21) printed pages including this page.

QUESTION 1 (20 MARKS)

- a) A 1.50 dm³ sample of a mixture of ethane gas, C₂H₆ and oxygen gas, measured at 25 °C and 101 kPa, was allowed to react in a bomb calorimeter in which, had a heat capacity of 5.03 kJ/K altogether with its contents. The complete combustion of the ethane gas to carbon dioxide gas and water caused a temperature rise in the calorimeter of 6.18 K. Given that ΔH_c° (C₂H₆) is -1560 kJ/mol.
- Define the standard enthalpy of combustion of ethane.
 - Write a thermochemical equation for the combustion of ethane.
 - Knowing the ΔH_c° of ethane, calculate the number of mole of ethane in the mixture.
 - Calculate the total moles of gases in the calorimeter.

(1+1+3+2 marks)

- b) Cyclohexene, C₆H₁₀ undergoes hydrogenation to produce cyclohexane, C₆H₁₂ according to the equation given below :



$$\Delta H_f^\circ (\text{C}_6\text{H}_{10}) = -30 \text{ kJ/mol}$$

$$\Delta H_f^\circ (\text{C}_6\text{H}_{12}) = -3930 \text{ kJ/mol}$$

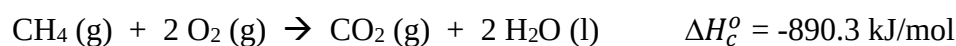
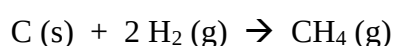
- Calculate the standard enthalpy of hydrogenation, ΔH° of cyclohexene.
- Sketch the enthalpy diagram of this reaction

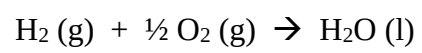
(2+2 marks)

- c) Butane is a hydrocarbon gas that commonly used for cooking. The standard enthalpy of combustion of butane is -220.1 kJ/mol. 10 g of butane is used to heat 8 L of water. Assuming that no heat is lost to the surroundings, what is the increase in the water's temperature? Given that the specific heat capacity of water is, $s(\text{H}_2\text{O}) = 4.18 \text{ J/g}^\circ\text{C}$ and water density is 1g/L.

(5 marks)

- d) Given the following data, determine the enthalpy change for the formation of 40 g of methane, CH₄.



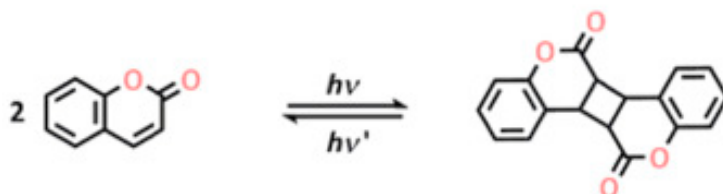


$$\Delta H_f^\circ = -285.8 \text{ kJ/mol}$$

(4 marks)

QUESTION 2 (10 MARKS)

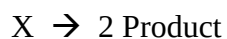
- a) Photodimerization is a chemical process when one photoexcited molecule reacts with one unexcited molecule of the same species. One example of this process is the reaction of coumarin as shown below.



- i. This reaction is reported as a second order reaction. Write a rate law equation for the forward reaction.
- ii. Based on your answer in a(i), calculate the initial concentration of coumarin, $\text{coumarin}]_0$ when k 0.137 L/mol.s and initial rate is found to be 3.4×10^{-4} mol/L.s

(1+3 marks)

- b) The experimental data for elimination reaction is shown below :



[X], (M)	Time (s)
0.100	0
0.067	50
0.050	100
0.040	150
0.033	200
0.029	250

Determine the rate law equation for the reaction using graphical method. Draw your graphs and state all your assumptions.

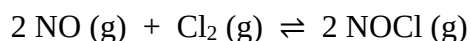
(6 marks)

QUESTION 3 (10 MARKS)

- a) 1 mole of N_2 and 3 mole of H_2 are placed in a flask at 400°C . Calculate the total pressure of the system at initial, P_0 in the function of x (the change) if the mole fraction of NH_3 is 0.43 at equilibrium.

(4 marks)

- b) Nitrosyl chloride, NOCl is produced from the reaction between nitric oxide gas and chlorine gas. At temperature $T^\circ\text{C}$, the partial pressures of NO , Cl_2 and NOCl in a 1 dm^3 equilibrium mixture are 60 kPa, 50 kPa and 80 kPa respectively.



- Calculate the value of equilibrium constant, K_p .
- When additional Cl_2 is added, its new partial pressure is 70 kPa. The new equilibrium partial pressure of NOCl is 88 kPa. Determine the partial pressure of NO and Cl_2 in the new equilibrium mixture if the temperature remains constant.

(1+5 marks)

QUESTION 4 (10 MARKS)

- a) Calculate the acid dissociation constant, K_a , of ammonium chloride, NH_4Cl , given the base dissociation constant, K_b , of ammonia is 2.6×10^{-4} .

(2 marks)

- b) Nitrous acid, HNO_2 is a weak acid which combine with X to form a buffer solution.

i. Give an example of X.

- ii. Calculate the amount (mole) of X to be used to be added to a beaker containing 50 mL of 0.6 M HNO_2 in order to form a buffer with pH of 4.5. Given the acid dissociation constant, K_a of HNO_2 is 7.2×10^{-4} .

(1+3 marks)

- c) Given that only 2.56 % of 1.0 M acetic acid, CH_3COOH , ionizes in water. Calculate the acid dissociation constant, K_a , of acetic acid.

(4 marks)

4) (a) $K_a \times K_b = K_w$

$$K_a = \frac{1 \times 10^{-14}}{2.6 \times 10^{-4}}$$

$$= 3.846 \times 10^{-11}$$

(b) weak acid: HNO_2

conjugate base: NO_2^-

salt of conjugate base: NaNO_2

$$\text{mol of } \text{HNO}_2 = 0.05 \text{ mol} \times 0.6$$

$$= 0.03$$

$$\text{pH} = \text{p}K_a + \log\left(\frac{[\text{c. base}]}{[\text{acid}]}\right)$$

$$4.5 = -\log(7.2 \times 10^{-4}) + \log\left(\frac{x}{0.03}\right)$$

$$x = 0.683 \text{ mol}$$

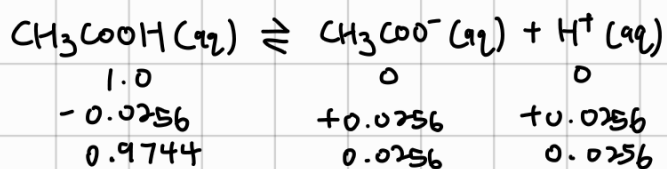
(c) $2.56\% = \frac{\text{mol of ionised}}{1.0 \text{ M}} \times 100$

$$= 0.0256 \text{ M}$$

$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]}$$

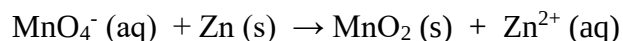
$$= \frac{0.0256^2}{0.9744}$$

$$= 6.726 \times 10^{-4}$$



QUESTION 5 (20 MARKS)

a) Given the redox reaction equation below:



- i. Complete and balance the above equation in basic solution.
- ii. Calculate the standard potential produced from the above redox reaction.
- iii. Write the Q_c expression for the above reaction

(5+2+1 marks)

b) Consider a concentration cell (voltaic cell) at temperature of 25°C. The setup consists of copper electrodes on both sides. One side is filled with 300 mL of 0.5 M copper sulfate, CuSO_4 , solution and the other side is 300 mL of 2.1 M CuSO_4 solution.

- i. Identify which concentration of CuSO_4 should be on cathode and anode.
- ii. Calculate the voltage produced by the cell.

(1+4 marks)

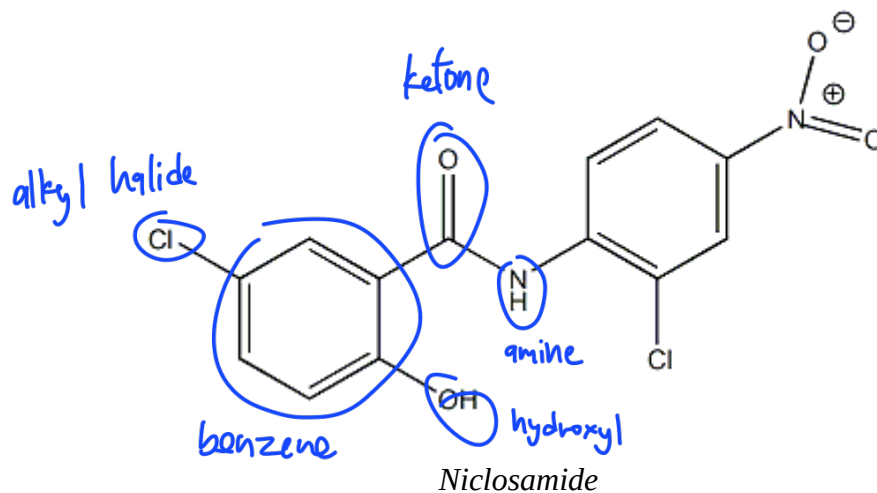
c) Consider 0.2 M of cobalt chloride, CoCl_2 , in an electrolytic cell with graphite electrodes being used on both sides.

- i. Write all the possible chemical equations that may occur on both anode and cathode.
- ii. Based on your answer in c(i), identify the major product that will be formed at the anode and cathode
- iii. If 2.2 A of current is applied on the cell for 30 minutes, calculate the mass of product formed at the cathode.

(2+2+3 marks)

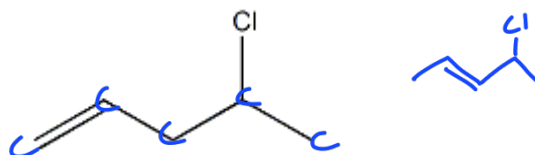
QUESTION 6 (10 MARKS)

- a) The structural formula of niclosamide, an-antiviral drugs which is currently use in clinical trial for Covid-19 cure is shown below. Draw and name TWO (2) functional groups from the structure.



(2 marks)

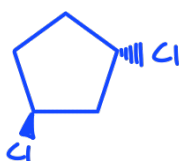
- b) The following structure is one isomer of C₅H₉Cl. Draw ONE (1) isomer of C₅H₉Cl that does not contain any primary carbon.



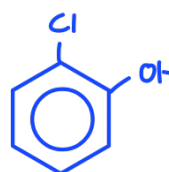
(2 marks)

- c) Draw the structure of the following molecules :

i. trans-1,3-Dichlorocyclopentane



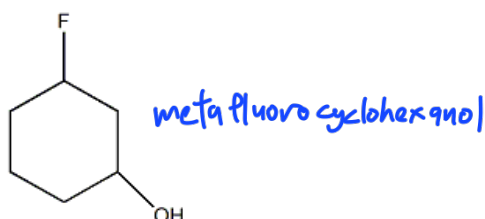
ii. o-Chlorophenol



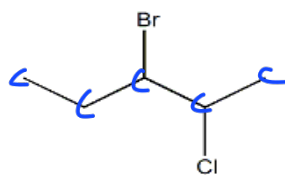
(1+1 marks)

- d) Give the IUPAC name for the following structures :

i.



ii.

*3-bromo-2-chloropentane*

(1+1 marks)

e) Draw TWO (2) cyclic ether structures from the chemical formula C_5H_8O .

(2 marks)

The Periodic Table

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
IA	IIA	IIIB	IVB	VB	VIB	VII B	VIII B			IB	II B	IIIA	IVA	VA	VIA	VIIA	VIIIA
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	2
H	He	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
1.01	4.00	10.81	12.01	14.01	16.00	19.00	20.18	22.99	24.31	26.98	28.09	30.97	32.07	35.45	39.95	39.1	40.08
3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
6.94	9.01	10.81	12.01	14.01	16.00	19.00	20.18	22.99	24.31	26.98	28.09	30.97	32.07	35.45	39.95	39.1	40.08
11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni
22.99	24.31	26.98	28.09	30.97	32.07	35.45	39.95	39.1	40.08	44.96	47.88	50.94	52.00	54.94	55.85	58.93	58.69
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
85.47	87.62	88.91	91.22	92.91	95.94	(98)	101.07	102.91	106.42	107.87	112.41	114.82	118.71	121.76	127.6	126.9	131.29
55	56	57	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
Cs	Ba	La*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
132.9	137.3	138.9	178.5	180.9	183.9	186.2	190.2	192.2	195.1	197.0	200.6	204.4	207.2	209	(209)	(210)	(222)
87	88	89	104	105	106	107	108	109	110	111							
Fr	Ra	Ac^	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg							
(223)	(226)	(227)	(261)	(262)	(263)	(264)	(265)	(268)	(271)	(272)							

List of Selected Constant Values

Speed of light in a vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$
Avogadro's number	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Faraday constant	F	=	$9.65 \times 10^4 \text{ C mol}^{-1}$
Planck constant	h	=	$6.6256 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	=	$1.054 \times 10^{-34} \text{ J s}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$ = $2.18 \times 10^{-14} \text{ J}$
Molar of gases constant	R	=	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ = $0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	k	=	$1.3807 \times 10^{-27} \text{ J K}^{-1}$
Mass of proton	M_p	=	$1.672 \times 10^{-27} \text{ kg}$
Electronic Bohr magneton	m_B	=	$9.2741 \times 10^{-24} \text{ J T}^{-1}$
Nuclear Bohr magneton	b_N	=	$5.05 \times 10^{-27} \text{ J T}^{-1}$
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	$1.602 \times 10^{-19} \text{ C}$

Unit and Conversion Factor

Energy	$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$ $1 \text{ calorie} = 4.184 \text{ Joule}$ $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Pressure	$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$

Half-Reaction	E_{red}°
$\text{K}^{+} + \text{e}^{-} \rightleftharpoons \text{K}$	-2.924
$\text{Ba}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Ba}$	-2.900
$\text{Ca}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Ca}$	-2.760
$\text{Na}^{+} + \text{e}^{-} \rightleftharpoons \text{Na}$	-2.712
$\text{Mg}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Mg}$	-2.375
$\text{H}_2 + 2\text{e}^{-} \rightleftharpoons 2\text{H}^{-}$	-2.230
$\text{Al}^{3+} + 3\text{e}^{-} \rightleftharpoons \text{Al}$	-1.706
$\text{Mn}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Mn}$	-1.040
$\text{Zn}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Zn}$	-0.763
$\text{Cr}^{3+} + 3\text{e}^{-} \rightleftharpoons \text{Cr}$	-0.740
$\text{S} + 2\text{e}^{-} \rightleftharpoons \text{S}^{2-}$	-0.508
$2\text{CO}_2 + 2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{H}_2\text{C}_2\text{O}_4$	-0.490
$\text{Cr}^{3+} + \text{e}^{-} \rightleftharpoons \text{Cr}^{2+}$	-0.410
$\text{Fe}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Fe}$	-0.409
$\text{Co}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Co}$	-0.280
$\text{Ni}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Ni}$	-0.230
$\text{Sn}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Sn}$	-0.136
$\text{Pb}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Pb}$	-0.126
$\text{Fe}^{3+} + 3\text{e}^{-} \rightleftharpoons \text{Fe}$	-0.036
$2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{H}_2$	0.000
$\text{S}_4\text{O}_6^{2-} + 2\text{e}^{-} \rightleftharpoons 2\text{S}_2\text{O}_3^{2-}$	0.089
$\text{Sn}^{4+} + 2\text{e}^{-} \rightleftharpoons \text{Sn}^{2+}$	0.150
$\text{Cu}^{2+} + \text{e}^{-} \rightleftharpoons \text{Cu}^{+}$	0.158
$\text{Cu}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Cu}$	0.340
$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^{-} \rightleftharpoons 4\text{OH}^{-}$	0.401
$\text{Cu}^{+} + \text{e}^{-} \rightleftharpoons \text{Cu}$	0.522
$\text{I}_3^{-} + 2\text{e}^{-} \rightleftharpoons 3\text{I}^{-}$	0.534
$\text{MnO}_4^{-} + 2\text{H}_2\text{O} + 3\text{e}^{-} \rightleftharpoons \text{MnO}_2 + 4\text{OH}^{-}$	0.588
$\text{O}_2 + 2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{H}_2\text{O}_2$	0.682
$\text{Fe}^{3+} + \text{e}^{-} \rightleftharpoons \text{Fe}^{2+}$	0.770
$\text{Hg}_2^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Hg}$	0.796
$\text{Ag}^{+} + \text{e}^{-} \rightleftharpoons \text{Ag}$	0.800
$\text{Hg}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Hg}$	0.851
$\text{H}_2\text{O}_2 + 2\text{e}^{-} \rightleftharpoons 2\text{OH}^{-}$	0.880
$\text{HNO}_3 + 3\text{H}^{+} + 3\text{e}^{-} \rightleftharpoons \text{NO} + 2\text{H}_2\text{O}$	0.960
$\text{Br}_2(\text{aq}) + 2\text{e}^{-} \rightleftharpoons 2\text{Br}^{-}$	1.087
$2\text{IO}_3^{-} + 12\text{H}^{+} + 10\text{e}^{-} \rightleftharpoons \text{I}_2 + 6\text{H}_2\text{O}$	1.190
$\text{CrO}_4^{2-} + 8\text{H}^{+} + 3\text{e}^{-} \rightleftharpoons \text{Cr}^{3+} + 4\text{H}_2\text{O}$	1.195
$\text{Pt}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Pt}$	1.200
$\text{MnO}_2 + 4\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{Mn}^{2+} + 2\text{H}_2\text{O}$	1.208
$\text{O}_2 + 4\text{H}^{+} + 4\text{e}^{-} \rightleftharpoons 2\text{H}_2\text{O}$	1.229
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^{+} + 6\text{e}^{-} \rightleftharpoons 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	1.330
$\text{Cl}_2(\text{g}) + 2\text{e}^{-} \rightleftharpoons 2\text{Cl}^{-}$	1.358
$\text{PbO}_2 + 4\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{Pb}^{2+} + 2\text{H}_2\text{O}$	1.467
$\text{MnO}_4^{-} + 8\text{H}^{+} + 5\text{e}^{-} \rightleftharpoons \text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.491
$\text{Au}^{+} + \text{e}^{-} \rightleftharpoons \text{Au}$	1.680
$\text{H}_2\text{O}_2 + 2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons 2\text{H}_2\text{O}$	1.776
$\text{Co}^{3+} + \text{e}^{-} \rightleftharpoons \text{Co}^{2+}$	1.842
$\text{S}_2\text{O}_8^{2-} + 2\text{e}^{-} \rightleftharpoons 2\text{SO}_4^{2-}$	2.050
$\text{O}_3(\text{g}) + 2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{O}_2(\text{g}) + \text{H}_2\text{O}$	2.070
$\text{F}_2(\text{g}) + 2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons 2\text{HF}(\text{aq})$	3.030

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